

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II EXAMINATION – WINTER 2024

Subject Code:BE01000031

Date:03-01-2025

Subject Name:Chemistry

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Explain de Broglie's concept in wave mechanics.	03
	(b) Explain the types of hybridization with the molecular shapes of the following compounds: CH ₄ and C ₂ H ₄ .	04
	(c) Discuss caustic embrittlement in boilers. Why does caustic embrittlement occur in boilers, and how can it be prevented?	07
Q.2	(a) Define the following terms:	03
	1. Octane Number	
	2. Fullerenes	
	3. Corrosion	
	(b) Compare scale formation and sludge formation in boilers.	04
	(c) Explain the deionization method used for softening of water with appropriate diagrams.	07
	OR	
	(c) Discuss the manufacturing of acetic acid through fermentation with appropriate diagrams.	07
Q.3	(a) What is meant by the mobile phase and stationary phase in liquid chromatography?	03
	(b) Differentiate between internal and external treatment for water softening.	04
	(c) Explain electrochemical theory of corrosion with appropriate diagrams.	07
	OR	
Q.3	(a) State the products formed by the action of the enzyme invertase and zymase on cane sugar under suitable fermentation conditions, along with the reactions.	03
	(b) Compare hard acids and bases with soft acids and bases.	04
	(c) What do you mean by spectroscopy? Explain the principle of NMR spectroscopy.	07

- | | | | |
|------------|------------|---|-----------|
| Q.4 | (a) | Briefly explain thin-layer chromatography. | 03 |
| | (b) | Discuss the applications of nanomaterials. | 04 |
| | (c) | Explain the principle and working of lithium batteries, including the reactions occurring in them. Reflect how they are better than other types of batteries. | 07 |
| OR | | | |
| Q.4 | (a) | List the applications of IR spectroscopy. | 03 |
| | (b) | What are inhibitors? Provide two examples. | 04 |
| | (c) | Analyze the structure of a bomb calorimeter and explain whether its design is suitable for measuring the calorific value. | 07 |
| Q.5 | (a) | Illustrate the uses of steel as an alloy with examples. | 03 |
| | (b) | Write about Heisenberg's Uncertainty Principle. | 04 |
| | (c) | Discuss the "Top-Down" and "Bottom-Up" approaches for synthesizing a nanomaterial. | 07 |
| OR | | | |
| Q.5 | (a) | Explain the sacrificial protection method for preventing metal corrosion with suitable diagrams. | 03 |
| | (b) | Discuss carbon nanotubes and nanowires in detail. | 04 |
| | (c) | Explain the refining of petroleum with the help of suitable diagrams. | 07 |

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I&II EXAMINATION – SUMMER 2025

Subject Code:BE01000031

Date:04-06-2025

Subject Name:Chemistry

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) What do you understand by biodegradable polymer?	03
	(b) Distinguish between chemical and electrochemical corrosion.	04
	(c) Describe with neat diagram of zeolite process and write its advantages and disadvantages.	07
Q.2	(a) Write the short note on caustic embrittlement.	03
	(b) Explain Top down and Bottom up approaches for synthesis of nanomaterial.	04
	(c) Discuss about pitting corrosion with neat diagram and mechanism.	07
	OR	
	(c) What is NMR spectroscopy? Explain with instrumentation and applications.	07
Q.3	(a) Write the applications of IR spectroscopy.	03
	(b) Explain properties and uses of copper, aluminum and nickel alloys.	04
	(c) Describe types of hybridization with suitable example of each type.	07
	OR	
Q.3	(a) What do you meant by biomass briquettes?	03
	(b) State Lambert and Beer's law and deduce its mathematical expression.	04
	(c) What is ferrous alloy? Explain heat treatments and applications of steel alloy.	07
Q.4	(a) Explain types of inhibitors with example.	03
	(b) Define and write Schrodinger's wave equation.	04
	(c) Describe with neat diagram of fractional distillation of crude petroleum.	07
	OR	
Q.4	(a) Write the properties and applications of fullerenes.	03
	(b) Distinguish between sludge and scale formation with examples.	04
	(c) What is chromatography? Explain principle and instrumentation of gas chromatography.	07
Q.5	(a) Define and write Heisenberg's Uncertainty principle.	03
	(b) Differentiate between proximate and ultimate analysis of coal.	04
	(c) Describe with suitable diagram to determination of calorific value by bomb calorimeter.	07
	OR	
Q.5	(a) Write the uses of carbon nanotubes and nanowires.	03
	(b) Explain the fermentation process for preparation of acetic acid.	04
	(c) Discuss about construction, working and uses of lithium batteries.	07

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: BE01000031/BE01R00031

Date: 20-12-2025

Subject Name: Chemistry

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Write short notes on hybridization.	03
(b) Write properties and applications of alloys.	04
(c) What is hard water? How it is affecting industries? Write the remediation for the same.	07
Q.2 (a) Write difference between: Scale & Sludge & Priming & Foaming	03
(b) Give details of Schrodinger's wave equation.	04
(c) How badly a metal can be affected by oxidation via moist air? Give definition types and two methods of prevention of corrosion.	07
OR	
(c) How will you solve the problem associated with efficiency of fuel? Explain your answer with respect to altering important properties of fuels.	07
Q.3 (a) Write notes on Heisenberg's uncertainty principle.	03
(b) How acidity and basicity is defined/varies in Hard and soft acid base concept.	04
(c) Write detailed notes on classification of alloys and heat treatment of steel.	07
OR	
(a) What is caustic embrittlement? Write chemistry behind it and remediation.	03
(b) Give two examples of each sp^3 and sp^3d type of hybridization.	04
(c) Describe energy storage devices with lithium batteries construction and working	07
Q.4 (a) What are Eco friendly materials? How they provide sustainable solution to pollution, give answer in context to bio based materials.	03
(b) Write short notes on: Red Shift, Retention factor, Hook's law, Shielding	04
(c) Write short notes on: Bio fuel, Bio diesel, Bio briquettes, photolysis of water, Knocking, octane number, calorific value	07

OR

- (a) Write applications of chemistry in our daily life, covering the topics associated with this course. 03
- (b) "Structural elucidation is incomplete without spectroscopic techniques", defend the statement with facts and figure. 04
- (c) What is Fermentation? Write its applications in synthesis of one very important chemical used in chemical industry (with detailed process, diagram and reaction). 07
- Q.5** (a) Breaking form top to bottom or making from bottom to top? Write on approaches used to make nano particles. 03
- (b) Write detailed note on GC. 04
- (c) Write the principle behind UV spectroscopy, draw its instrumentation diagram and elaborate application. 07

OR

- (a) Write a note on: metal nano particles, fullerenes, carbon nano tubes 03
- (b) What is chromatography? Why it is important to do TLC in basic synthesis in chemistry laboratory? 04
- (c) Nikita synthesized o-nitro phenol in chemistry laboratory. Help her analyzing the same with the help of UV, IR, NMR spectroscopy with tentative peak assignments and evaluation of the structure. 07
